Fiscal Year:	FY 2024	**Task Last Updated:**	FY 09/08/2023
PI Name:	Weber, Richard Ph.D.		
Project Title:	Microgravity Investigation of Thermophysical Properties of Supercooled Molten Metal Oxides		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
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City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/21/2017	**End Date:**	05/20/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 05/20/2025 per NSSC information (Ed., 6/21/23) NOTE: End date changed to 05/20/2024 per NSSC information (Ed., 6/21/23) NOTE: End date changed to 05/20/2023 per NSSC information (Ed., 8/25/22) NOTE: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2022; added to original grant since is continuation of the work (Ed., 5/18/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0059 ; NNX17AH73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is investigating the thermophysical properties of molten metal oxides. These materials are of critical importance in a wide range of applications including functional glasses, batteries, next generation propulsion systems, lasers and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. Fundamental research provides an essential underpinning for transformational discovery and development of major new applications. This project aims to measure the density, thermal expansion coefficient, viscosity, and surface tension of a variety of molten oxides, some of which can form glass. Containerless conditions are essential in order to achieve deep supercooling without extrinsic heterogeneous nucleation of crystals. This requirement will be met by using the Japanese Aerospace Exploration Agency (JAXA) Electrostatic Levitation Furnace (ELF). The ELF instrument is an electrostatic levitator designed for work on metal oxides at temperatures up to about 2500 ºC in containerless conditions. By working in microgravity, nearly quiescent molten samples can be obtained. Induced oscillations of levitated liquid droplets is used to study their fluid properties. Image analysis reveals the oscillation amplitude and power spectrum, from which the required properties (surface tension and viscosity) are calculated. The materials of primary interest are molten metal oxides, many of which form fragile liquids that can be made into glass provided the melts can be deeply supercooled. In some cases there is already considerable data on the atomic structure of these melts. An important goal of this research is to correlate the structure and properties (in particular viscosity and density) to advance models of the processing of materials. These models are important in advancing basic science of liquids and glasses and in development of high value-added applied materials. Samples returned from the flight experiments are being investigated by a variety of techniques and in collaboration with several research groups at US universities and National Laboratories. ED. NOTE 5/18/2020: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2024; added to original grant since is continuation of the work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is investigating molten metal oxides. These materials are of critical importance in a wide range of applications including functional glasses, batteries, next generation propulsion systems, lasers and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. Fundamental research provides an essential underpinning for transformational discovery and development of major new applications. The project work includes outreach to undergraduate students. The experiments in microgravity are measuring essential thermophysical property data that can be used to advance understanding of how to process and optimize materials performance.
	NASA Grant Number 80NSSC18K0059 (Formerly NNX17AH73G) Microgravity Investigation of Thermophysical Properties of Supercooled Molten Metals Oxides (“SuperGlass”) Annual Progress Report Grant period: 11/09/2017-5/20/2024 Reporting period: 11/20/2022-11/10/2023 Richard Weber Materials Development, Inc., 3090 Daniels Court, Arlington Heights, IL 60004 This project is investigating the thermophysical properties of molten metal oxides. These materials are of critical importance in a wide range of applications, including functional glasses, batteries, next-generation propulsion systems, lasers, and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value to technologies for processing planetary materials in space exploration. Fundamental research provides an essential underpinning for transformational discovery and the development of major new applications. This project is measuring the density, thermal expansion coefficient, viscosity, and surface tension of a variety of molten oxides, some of which can form glass. Containerless conditions are essential to achieve deep supercooling without extrinsic heterogeneous nucleation of crystals. This requirement will be met by using the Japanese Aerospace Exploration Agency (JAXA) Electrostatic Levitation Furnace (ELF). The ELF instrument is an electrostatic levitator designed for work on metal oxides at temperatures up to about 2500 ºC in containerless conditions. By working in microgravity, nearly quiescent molten samples can be obtained. Induced oscillations of levitated liquid droplets are used to study their fluid properties. Image analysis reveals the oscillation amplitude and power spectrum, from which the required properties (surface tension and viscosity) are calculated. The materials of primary interest are molten metal oxides, many of which form fragile liquids that can be made into glass, provided the melts can be deeply supercooled. In some cases, there is already considerable data on the atomic structure of these melts. An important goal of this research is to correlate the structure and properties (in particular, viscosity and density) to advance models for the processing of materials. These models are important in advancing basic science of liquids and glasses and in the development of high value-added applied materials. Samples returned from the flight experiments are being investigated by a variety of techniques and in collaboration with several research groups at U.S. universities and national laboratories. 1. Status: The project team includes Materials Development Inc. (MDI) scientists and Professor Robert Hyers, University of Massachusetts, Amherst; Dr. Shinji Kohara, National Institute for Materials Science, Japan; and Dr. Takehiko Ishikawa, JAXA, Japan. Experiments on ELF are performed by Dr. Chihiro Koyama and coordinated by Mr. Hirohisa Oda from JAXA. The collaborators from Japan are not funded by this grant. The grant was awarded on 5/21/2017 and assigned a new contract number in November 2017. First-year funding was at a level sufficient to scope the work and perform preliminary ground-based experiments in preparation for a more detailed experimental campaign during the subsequent years. Second-year funding was at a level sufficient for planning and preparation for the Science Concept Review (SCR). The SCR was successfully completed in March 2020, and the remaining issues from the SCR were addressed soon afterward. The first round of flight experiments was started in November 2021 and completed in March 2022. This work included experiments on more than 30 samples. Due to some operational issues, a few samples could not be run. In an effort to complete the investigation, JAXA offered to provide a fourth sample holder, and MDI has made preparations for the additional samples. The fourth sample holder was sent to the International Space Station (ISS) on SpX-27 in March 2023, and experiments are planned for early 2024. MDI was granted an extension of the period of performance due to delays in the implementation of ISS experiments. 2. Technical Activities: Activities during the sixth year included: (i) analyzing data and recovered samples, (ii) making ground-based structure and property measurements on glass and liquid samples, (iii) delivering additional samples to JAXA that were launched in March 2023, and (iv) publishing results. Starting in August 2022, the MDI team is holding

Task Progress:	bi-weekly meetings with Brandon Phillips at NASA Marshall Space Flight Center to discuss progress and plan future activities. (i) Flight experiments were run in ELF during the period November 2021-March 2022. Five samples each of nine different compositions were launched. A precision stainless steel calibration sphere (identified as SUS) was measured at the start and end of each sample holder. Data analysis is continuing. The methodology for calculating sample volume compared to the calibration sphere was refined, based on a collaboration with Professor Robert Hyers. Following these modifications to the density analysis algorithm, all sample density analyses were re-run, using mass data from recovered samples and internal porosity corrections obtained from post-flight X-ray tomography. Viscosity and surface tension data were obtained from the drop oscillation spectra. These data are being reduced using programs provided by JAXA. Work is in progress to use theLarge-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) molecular dynamics package to model the liquid samples. MDI is collaborating with Professor Collin Wilkinson at Alfred University. He is an expert in modeling melt properties from first principles. Comparing modeled data with measured structure factors will provide a check on the quality of the model fit, lending additional weight to the value of the data. The model will then be used for calculating viscosity, to compare with the ELF flight measurements. (ii) Ground-based tests are in progress to characterize the compositions for flight and to make measurements on the samples recovered from the flight experiments. Flight samples were examined by optical microscopy to determine the overall appearance of the beads. High energy white beam X-ray tomography was performed at the Advanced Photon Source beamline 7 BM-B to determine the interior structure of the samples with a resolution of ca. 5 µm. These measurements are useful to evaluate the presence of bubbles or interior features that might affect the calculated volume (and hence density) of glass samples. In all cases, both flight and ground-based samples were examined to enable comparison of the materials processed in microgravity and on Earth. Glass samples were recently returned from the Spallation Neutron Source where additional structure measurements were made. These samples are being sent to the Materials of the Universe Laboratory at Arizona State University. There, the enthalpy of the solution will be measured to obtain the heat of vitrification. For the YbAG composition, the atomic structure of the liquid was measured using high energy x-ray diffraction at the Advanced Photon Source, Sector 6-ID-D, and isotope substitution neutron diffraction at the Spallation Neutron Source. (iii) Samples for the fourth holder were supplied to JAXA in August 2022. Several samples are doped with 1% Fe2O3 (“Fe1”) to improve absorption of the heating laser. The holder was launched on SpX-27 in March 2023. Flight experiments using the ELF instrument are currently planned for early 2024. Two pure aluminum oxide samples are included. These samples will be used to calibrate the temperature measurements using the well-established melting point of aluminum oxide as a reference. The expected oscillation frequencies of molten samples were estimated using literature data for surface tension. Data were typically not available for the exact compositions under study, so the values were obtained by extrapolating or interpolating data for similar materials from the literature [Ed. Note: See References 1-5]. The estimated oscillation frequencies provide a useful starting point for measurements in ELF. The initial coarse scan of frequencies used to obtain the resonance frequency of a liquid drop will be based on bracketing the estimated surface tension by +/- 100 mN/m. The typical frequency range is from about 120-160 Hz. (iv) Presenting and publishing results is an important aspect of the research. Details of publications are given below. [Ed. Note: See also the Cumulative Bibliography.] 3. Publications/presentations: Dr. Stephen Wilke presented an invited lecture for the National Institute of Standards and Technology (NIST) Center for Neutron Research in November 2022. Papers were presented by Dr. Stephen Wilke (1 invited, 1 contributed) and Dr. Richard Weber (1 contributed) at the American Ceramic Society Glass and Optical Materials Division meeting, June 4-8, 2023. Dr. Weber presented an invited talk at the 36th Fall Meeting of the Ceramic Society of Japan (via video link). Papers will be presented at the ASGSR annual meeting, in Washington, DC, November 13-18, 2023. Two peer-reviewed scientific papers were published: -- S.K. Wilke, C.J. Benmore, J. Ilavsky, R.E. Youngman, A. Rezikyan, M.P. Carson, V. Menon and R. Weber, “Phase separation in mullite-composition glass,” Sci. Rep., 12, 17687 (2022). -- S.K. Wilke, C.J. Benmore, J. Ilavsky, R.E. Youngman, A. Rezikyan, M.P. Carson, V. Menon and R. Weber, “Revisiting metastable immiscibility in SiO2–Al2O3: Structure and phase separation of supercooled liquids and glasses,” J. Am. Ceram. Soc., 106, 2820 (2023). Editor’s Choice paper. -- A paper on the flight data is in development. 4. Plans: Plans for the next project period are: (i) Continue data analysis, (ii) Continue sample analysis and characterization, (iii) Run ISS-ELF experiments on samples in the fourth holder, (iv) Visit NASA Marshall Space Flight Center to perform ground-based experiments using the hybrid levitator, (v) Publish/present results. 5. References [1] Kingery, J. Am. Ceram. Soc., 42, 6, (1959). [2] Bates, et al., Contract report NAS 3-14308, (1972). [3] Choi, et al., ISIJ International, 42, 221 (2002). [4] Weirauch, et al., J. Am. Ceram. Soc., 79, 920 (1996). [5] Conte, et al., Materials, 11, 2475 (2018).
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Wilke SK, Benmore CJ, Ilavsky J, Youngman RE, Rezikyan A, Carson MP, Menon V, Weber R. "Phase separation in mullite-composition glass." Scientific Reports. 2022 Oct 21;12(1):17687. https://doi.org/10.1038/s41598-022-22557-7 , Oct-2022

Articles in Peer-reviewed Journals	Wilke SK, Benmore CJ, Menon V, Ilavsky J, Rezikyan A, Youngman RE, Carson MP, Weber R. "Revisiting metastable immiscibility in SiO2–Al2O3: Structure and phase separation of supercooled liquids and glasses. " J. Am. Ceram. Soc. 2023 May;106(5):2820-34. https://doi.org/10.1111/jace.18960 , May-2023
Articles in Peer-reviewed Journals	Alderman OL, Benmore CJ, Reynolds B, Royle B, Feller S, Weber RJ. "Liquid fragility maximum in lithium borate glass-forming melts related to the local structure." International Journal of Applied Glass Science. 2023 Jan;14(1):52-68. https://doi.org/10.1111/ijag.16611 , Jan-2023